

## 173. Physical Medium Attachment (PMA) sublayer, type 800GBASE-R

### 173.1 Overview

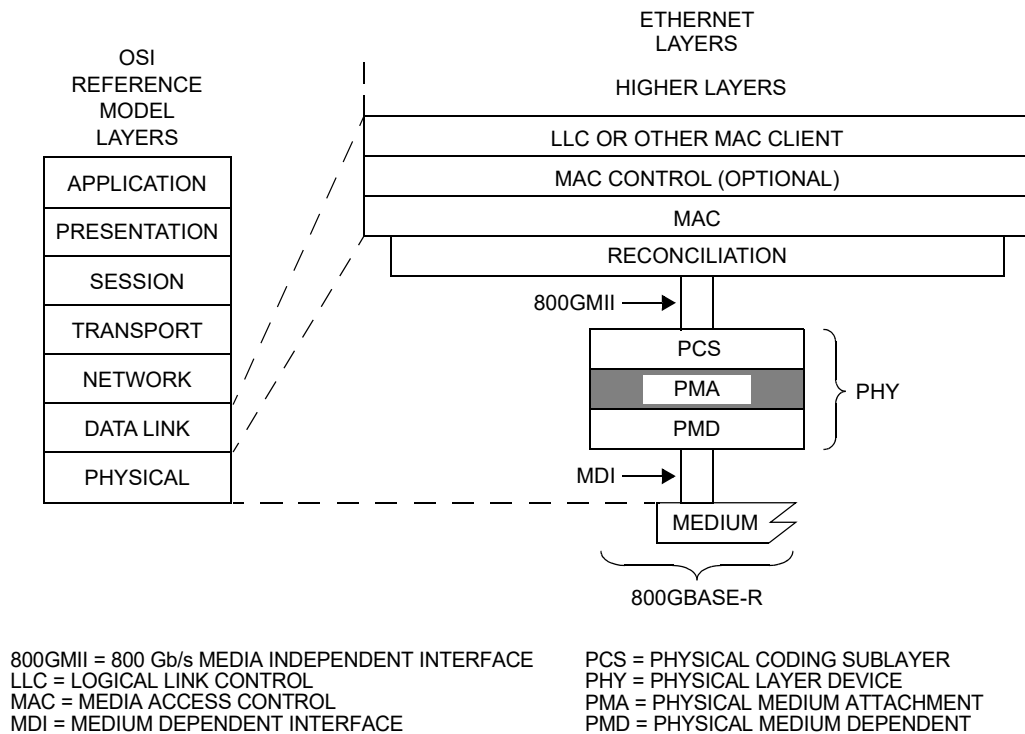
#### 173.1.1 Scope

This clause specifies the Physical Medium Attachment (PMA) sublayer for the family of 800 Gb/s Physical Layer implementations, known as 800GBASE-R. The PMA allows the PCS (specified in Clause 172) to connect in a media-independent way with a range of physical media, or for the DTE 800GXS to connect to the PHY 800GXS (specified in Clause 171). The 800GBASE-R PMA(s) can support any of the 800 Gb/s PMDs in Table 169–1.

Three types of the 800GBASE-R PMA are defined: 32:8 PMA, 8:32 PMA, and 8:8 PMA.

#### 173.1.2 Position of the PMA in the 800GBASE-R sublayers

Figure 173–1 shows the relationship of the PMA sublayer with other sublayers to the ISO Open System Interconnection (OSI) reference model.



**Figure 173–1—800GBASE-R PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model**

### 173.1.3 Summary of functions

The following is a summary of the principal functions implemented (when required) by the PMA in both the transmit and receive directions:

- Adapt the PCS lane (PCSL) formatted signal to the appropriate number of abstract or physical lanes
- Provide per input-lane clock and data recovery
- Provide bit-level multiplexing
- Provide clock generation
- Provide signal drivers
- Optionally provide local loopback to/from the PMA service interface
- Optionally provide remote loopback to/from the PMD service interface
- Optionally provide test-pattern generation and detection
- Tolerate Skew Variation
- Perform PAM4 encoding and decoding
- Provide signal status information

### 173.1.4 PMA sublayer positioning

An implementation may use one or more PMA sublayers to adapt the number and rate of the PCSs to the number and rate of the PMD lanes. The number of PMA sublayers required depends on the partitioning of functionality for a particular implementation.

Figure 173–2 shows examples of the PMA sublayer positioning for implementations with an 800GMII Extender and for implementations without an 800GMII Extender. Additional examples are illustrated in Annex 173A.

A 32:8 PMA sublayer is required in the following cases:

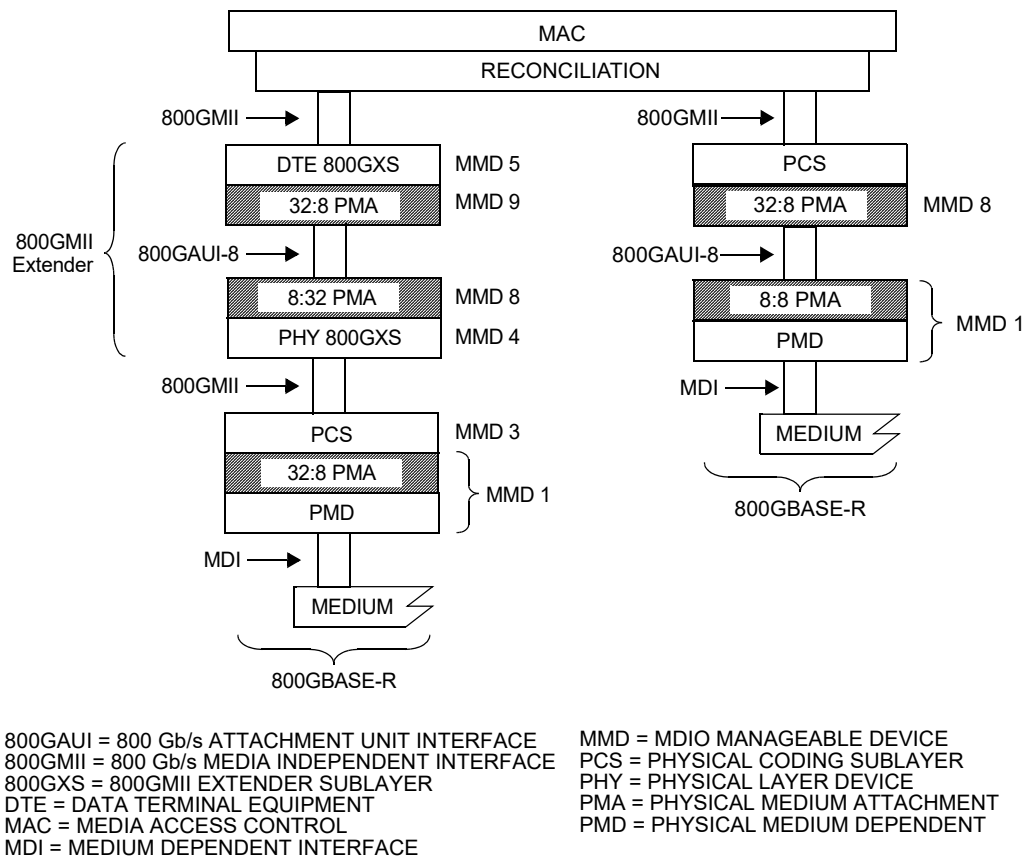
- In an 800GBASE-R PHY between the 800GBASE-R PCS and an 800GAUI-8
- In an 800GBASE-R PHY between the 800GBASE-R PCS and an 8-lane 800GBASE-R PMD
- In an 800GMII Extender between the DTE 800GXS and an 800GAUI-8

An 8:32 PMA sublayer is required in the following case:

- In an 800GMII Extender between an 800GAUI-8 and the PHY 800GXS

An 8:8 PMA sublayer is required in the following cases:

- In an 800GBASE-R PHY between an 800GAUI-8 and an 8-lane 800GBASE-R PMD
- In an 800GBASE-R PHY between two 800GAUI-8
- In an 800GMII Extender between two 800GAUI-8



**Figure 173-2—Examples of 800GBASE-R PMA layering**

Management Data Input/Output (MDIO) Manageable Device (MMD) addresses 1, 8, 9, 10, and 11 are available for addressing multiple instances of PMA sublayers (see Table 45-1 for MMD device addresses). If the PMA sublayer that is closest to the PMD is packaged with the PMD, it shares MMD 1 with the PMD. More addressable instances of PMA sublayers, each one separated from lower addressable instances by chip-to-chip interfaces, may be implemented and addressed allocating MMD addresses to PMAs in increasing numerical order going from the PMD toward the MAC. A maximum of four PMA sublayers are addressable as MDIO MMDs.

## 173.2 PMA service interface

The PMA service interface for 800GBASE-R is an instance of the inter-sublayer service interface defined in 169.3.1. The PMA service interface primitives (when applicable) are summarized as follows:

```
PMA:IS_UNITDATA_i.request(tx_symbol)
PMA:IS_UNITDATA_i.indication(rx_symbol)
PMA:IS_SIGNAL.indication(SIGNAL_OK)
```

For a 32:8 PMA (see Figure 173-3), the primitives are defined for  $i = 0$  to 31. In the transmit direction, the PMA receives 32 parallel bit streams from either the 800GBASE-R PCS or the DTE 800GXS, each at a nominal signaling rate of 26.5625 Gb/s. In the receive direction, the PMA sends 32 parallel bit streams to the PMA client, each at a nominal signaling rate of 26.5625 Gb/s. The 32:8 PMA also provides signal status

information to the PMA client using the PMA:IS\_SIGNAL.indication service interface primitive (see 173.5.8.1).

For the 8:32 and 8:8 PMAs (see Figure 173–4 and Figure 173–5), the service interface primitives are defined for  $i = 0$  to 7. In the transmit direction, the PMA receives 8 parallel PAM4 symbol streams from the PMA client, each operating at a nominal signaling rate of 53.125 GBd. In the receive direction, the PMA sends 8 parallel PAM4 symbol streams to the PMA client, each at a nominal signaling rate of 53.125 GBd.

The 8:32 and 8:8 PMAs may optionally provide signal status information to the PMA client as described in 173.5.8.2 and 173.5.8.3.

### 173.3 Service interface below PMA

There are several different sublayers that may appear below a PMA, including the PMD, an Extender Sublayer, or another PMA. The variable *inst* represents whichever sublayer appears below the PMA (e.g., another PMA or a PMD).

The sublayer below the PMA utilizes the inter-sublayer service interface defined in 169.3.1. The service interface primitives provided to the PMA (when applicable) are summarized as follows:

```
inst:IS_UNITDATA_i.request(tx_symbol)
inst:IS_UNITDATA_i.indication(rx_symbol)
inst:IS_SIGNAL.indication(SIGNAL_OK)
PHY_XS:IS_SIGNAL.request(SIGNAL_OK)
```

For the 32:8 PMA and 8:8 PMA (see Figure 173–3 and Figure 173–5), the *inst:IS\_UNITDATA\_i* primitives are defined for  $i = 0$  to 7 and *inst* is either PMA or PMD. In the transmit direction, the PMA sends 8 parallel PAM4 symbol streams to the sublayer below the PMA, each operating at a nominal signaling rate of 53.125 GBd. In the receive direction, the PMA receives 8 parallel PAM4 symbol streams from the sublayer below the PMA, each at a nominal signaling rate of 53.125 GBd. Electrical and timing specifications of the service interface below the PMA are defined if the interface is physically instantiated (e.g., 800GAUI-8), otherwise the service interface is specified only abstractly. If the sublayer below the PMA is a PMD, then the PMA also receives a PMD:IS\_SIGNAL.indication primitive indicating the status of the signal from the PMD (see 173.5.8.1 and 173.5.8.3).

For the 8:8 PMA, if the sublayer below the PMA is another PMA, the 8:8 PMA may optionally provide signal status information to the sublayer below as described in 173.5.8.3.

For the 8:32 PMA (see Figure 173–4) the *inst:IS\_UNITDATA\_i* primitives are defined for  $i = 0$  to 31 and *inst* is always PHY\_XS. In the transmit direction, the PMA sends 32 parallel bit streams to the PHY 800GXS, each at a nominal signaling rate of 26.5625 Gb/s. In the receive direction, the PMA receives 32 parallel bit streams from the PHY 800GXS, each at a nominal signaling rate of 26.5625 Gb/s. The PMA provides signal status information to the PHY 800GXS using the PHY\_XS:IS\_SIGNAL.request service interface primitive (see 173.5.8.2). The PMA also receives a PHY\_XS:IS\_SIGNAL.indication primitive indicating the status of the signal from the PHY 800GXS (see 173.5.8.2).

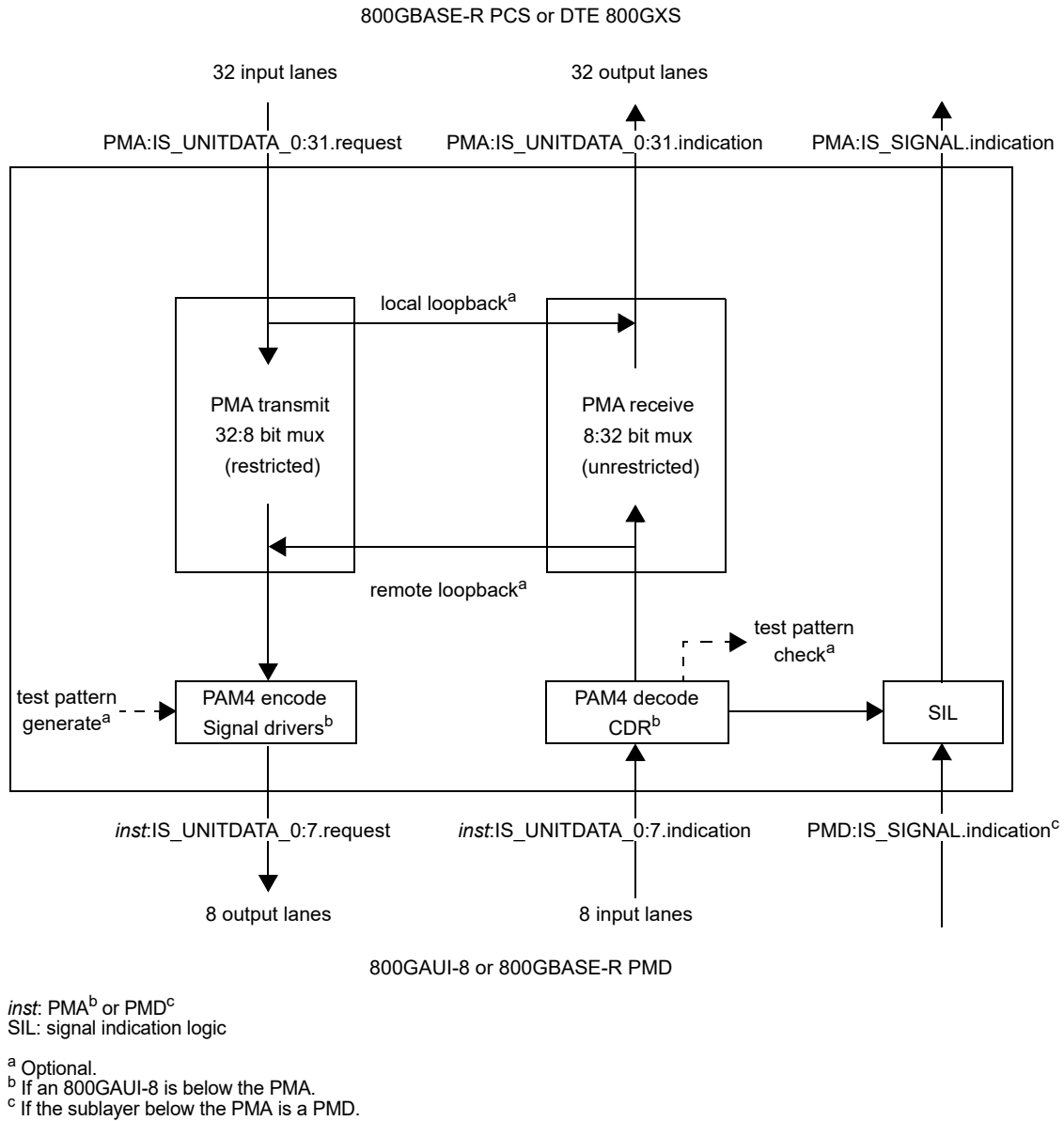
### 173.4 PMA types

Three different types of the 800GBASE-R PMA are defined in this clause: 32:8 PMA, 8:32 PMA, and 8:8 PMA. These PMA types are all based upon the 400GBASE-R PMA defined in Clause 120.

### 173.4.1 32:8 PMA

Figure 173–3 provides the functional block diagram of the 32:8 PMA. The features of a 32:8 PMA include the following:

- 32 input and output lanes for the PMA service interface
- 8 input and output lanes for the service interface below the PMA
- In the transmit direction, restricted bit-level multiplexing between the 32 input lanes and the 8 output lanes (see 173.5.2.1)
- In the receive direction, unrestricted bit-level multiplexing between the 8 input lanes and the 32 output lanes (see 173.5.2.1)
- The interface above the PMA (32 lanes) connects with either an 800GBASE-R PCS or a DTE 800GXS
- The interface below the PMA (8 lanes) connects with either a PMD or a physically instantiated interface (800GAUI-8)



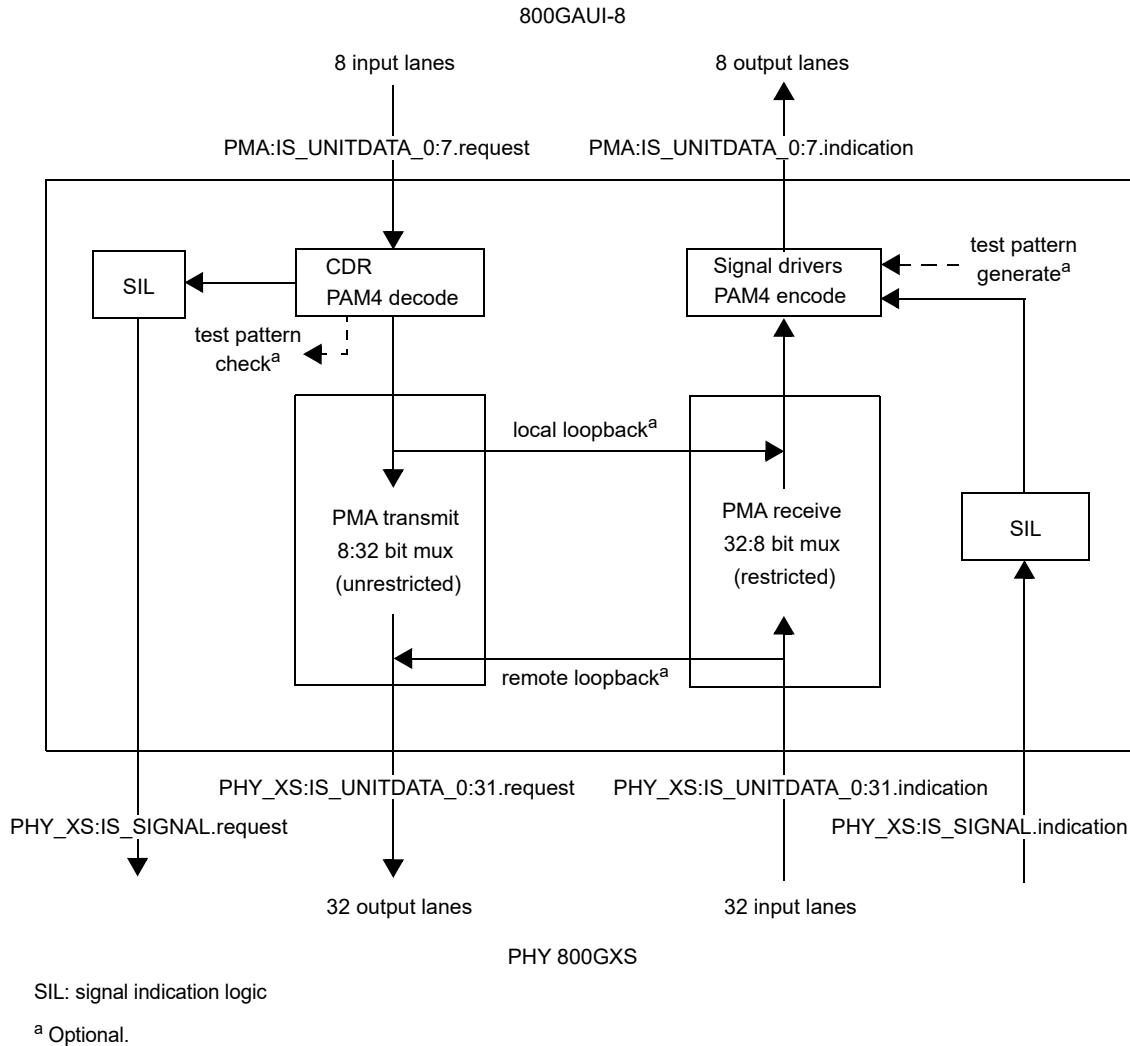
**Figure 173-3—32:8 PMA functional block diagram**

#### 173.4.2 8:32 PMA

Figure 173-4 provides the functional block diagram of the 8:32 PMA. The features of an 8:32 PMA include the following:

- 8 input and output lanes for the PMA service interface
- 32 input and output lanes for the service interface below the PMA
- In the transmit direction, unrestricted bit-level multiplexing between the 8 input lanes and the 32 output lanes (see 173.5.2.2)
- In the receive direction, restricted bit-level multiplexing between the 32 input lanes and the 8 output lanes (see 173.5.2.2)

- The interface above the PMA (8 lanes) connects with a physically instantiated interface (800GAUI-8)
- The interface below the PMA (32 lanes) connects with a PHY 800GXS

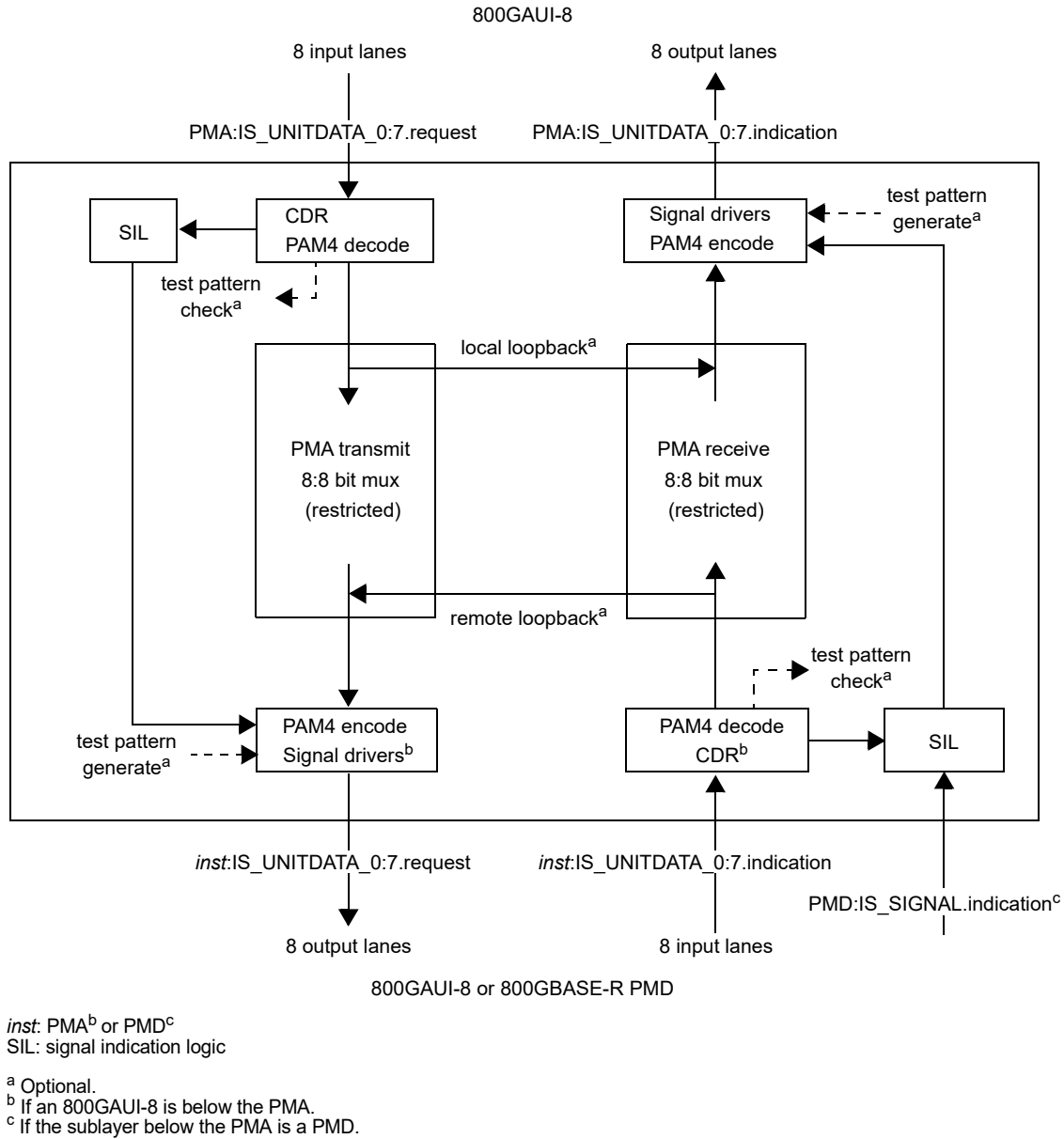


**Figure 173-4—8:32 PMA functional block diagram**

### 173.4.3 8:8 PMA

Figure 173-5 provides the functional block diagram of the 8:8 PMA. The features of an 8:8 PMA include the following:

- 8 input and output lanes for the PMA service interface
- 8 input and output lanes for the service interface below the PMA
- In the transmit and receive directions, restricted bit-level multiplexing between the 8 input lanes and the 8 output lanes (see 173.5.2.3)
- The interface above the PMA connects with a physically instantiated interface (800GAUI-8)
- The interface below the PMA connects with either a PMD or a physically instantiated interface (800GAUI-8)



**Figure 173-5—8:8 PMA functional block diagram**

## 173.5 Functions within the PMA

### 173.5.1 Per input-lane clock and data recovery (CDR)

If the interface above the PMA is physically instantiated as 800GAUI-8, the PMA shall meet the electrical and timing specifications in Annex 120F or Annex 120G, as appropriate, at the PMA service interface.

If the interface below the PMA is physically instantiated as 800GAUI-8, the PMA shall meet the electrical and timing specifications in Annex 120F or Annex 120G, as appropriate, at the service interface below the PMA.

### 173.5.2 Bit-level multiplexing

The PMA provides bit-level multiplexing of PCSs in both the transmit and receive directions.

The bit-level multiplexing functions are different amongst the 32:8, 8:32, and 8:8 PMAs.

#### 173.5.2.1 32:8 PMA bit-level multiplexing

The 32:8 PMA provides bit-level multiplexing in both the transmit and receive directions.

In the transmit direction, the bit-level multiplexing function is performed among the PCSs received from the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes  $i = 0$  to 31) with the result sent to the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes  $i = 0$  to 7), as shown in Figure 173–3. This is a restricted bit-level multiplexing function that is identical to the bit-level multiplexing function specified in 120.5.2 with the number of PCSs  $z$  equal to 32 and with the following exception:

- The bit-level multiplexing function has an additional constraint that each of the 8 output lanes carries two PCSs from PMA client lanes  $i = 0$  to 15 and two PCSs from PMA client lanes  $i = 16$  to 31. Bits from the four PCSs are multiplexed in temporal order with one bit from each of two lanes from PMA client lanes  $i = 0$  to 15 followed by one bit from each of two lanes from PMA client lanes  $i = 16$  to 31.

In the receive direction, the bit-level multiplexing function is performed among the PCSs received from the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes  $i = 0$  to 7) with the result sent to the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes  $i = 0$  to 31), as shown in Figure 173–3. This is an unrestricted bit-level multiplexing function that is identical to that specified in 120.5.2 with the number of PCSs  $z$  equal to 32.

#### 173.5.2.2 8:32 PMA bit-level multiplexing

The 8:32 PMA provides bit-level multiplexing in both the transmit and receive directions.

In the transmit direction, the bit-level multiplexing function is performed among the PCSs received from the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes  $i = 0$  to 7) with the result sent to the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes  $i = 0$  to 31), as shown in Figure 173–4. This is an unrestricted bit-level multiplexing function that is identical to the bit-level multiplexing function specified in 120.5.2 with the number of PCSs  $z$  equal to 32.

In the receive direction, the bit-level multiplexing function is performed among the PCSs received from the service interface below the PMA using the `PHY_XS:IS_UNITDATA_i.indication` primitives (for service interface lanes  $i = 0$  to 31) with the result sent to the PMA client via the `PMA:IS_UNITDATA_i.indication` primitives (for PMA client lanes  $i = 0$  to 7), as shown in Figure 173–4. This is a restricted bit-level multiplexing function that is identical to that specified in 120.5.2 with the number of PCSs  $z$  equal to 32 and with the following exception:

- The bit-level multiplexing function has an additional constraint that each of the 8 output lanes carries two PCSs from PHY 800GXS service interface lanes  $i = 0$  to 15 and two PCSs from PHY 800GXS service interface lanes  $i = 16$  to 31. Bits from the four PCSs are multiplexed in temporal order with one bit from each of two lanes from PHY 800GXS service interface lanes  $i = 0$  to 15 followed by one bit from each of two lanes from PHY 800GXS service interface lanes  $i = 16$  to 31.

### 173.5.2.3 8:8 PMA bit-level multiplexing

The 8:8 PMA provides bit-level multiplexing in both the transmit and receive directions.

In the transmit direction, the bit-level multiplexing function is performed among the PCSs received from the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes  $i = 0$  to 7) with the result sent to the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes  $i = 0$  to 7), as shown in Figure 173–5.

In the receive direction, the bit-level multiplexing function is performed among the PCSs received from the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes  $i = 0$  to 7) with the result sent to the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes  $i = 0$  to 7), as shown in Figure 173–5.

In both the transmit and receive directions, this is a restricted bit-level multiplexing function that is identical to the bit-level multiplexing function specified in 120.5.2 with the number of PCSs  $z$  equal to 32 and with the following exception:

- The 4 PCSs received on an input lane shall be mapped to an output lane such that the Gray mapped PAM4 symbol sequence on the output lane is either identical to the Gray mapped PAM4 symbol sequence on the input lane, or is the result of reversing the order of each pair of bits {A, B} to {B, A} in the Gray mapping function (see 173.5.7.1).

### 173.5.3 Skew and Skew Variation

The Skew (relative delay) between the PCSs is kept within limits so that the information on the lanes can be reassembled by the PCS.

Any PMA that combines PCSs from different input lanes onto the same output lane is required to tolerate Skew Variation between the input lanes without changing the PCSL positions on the output. Skew and Skew Variation at physically instantiated interfaces are specified at Skew points SP1 and SP2 in the transmit direction and SP5 and SP6 in the receive direction. Skew, Skew Variation, and Skew points are defined in 169.5. Skew points are illustrated in Figure 169–4 and Figure 169–5.

#### 173.5.3.1 Skew generation toward SP1

In an implementation with one or more physically instantiated 800GAUI-8 interfaces, at the output of the PMA that sends data in the transmit direction toward the 800GAUI-8 that is closest to the PMD (SP1 in Figure 169–4 and Figure 169–5) there shall be no more than 16 ns of Skew between PCSs and no more than 200 ps of Skew Variation between output lanes towards the 800GAUI-8.

#### 173.5.3.2 Skew tolerance at SP1

In an implementation with one or more physically instantiated 800GAUI-8 interfaces, the PMA that receives data in the transmit direction from the 800GAUI-8 (SP1 in Figure 169–4 and Figure 169–5) shall tolerate the maximum amount of Skew Variation allowed at SP1 (200 ps) between input lanes while maintaining the bit ordering and position of each PCSL on each PMA lane in the transmit direction (toward the PMD).

#### 173.5.3.3 Skew generation toward SP2

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, at the output of the PMA adjacent to the PMD service interface there shall be no more than 25 ns of Skew between PCSs, and no more than 400 ps of Skew Variation between output lanes toward the PMD service interface (SP2 in Figure 169–4 and Figure 169–5). In an implementation with one or more physically instantiated 800GAUI-8

interfaces, the Skew measured at the input to the PMA adjacent to the PMD service interface (SP1 in Figure 169–4 and Figure 169–5) is limited to no more than 16 ns of Skew and no more than 200 ps of Skew Variation.

#### 173.5.3.4 Skew tolerance at SP5

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, the PMA adjacent to the PMD service interface (SP5) shall tolerate the maximum amount of Skew Variation allowed at SP5 (3.6 ns) between output lanes from the PMD service interface while maintaining the bit ordering and position of each PCSL on each PMA lane in the receive direction (toward the MAC).

#### 173.5.3.5 Skew generation at SP6

In an implementation with one or more physically instantiated 800GAUI-8 interfaces, at SP6 (the receive direction of the 800GAUI-8 closest to the PCS), at the output of the PMA or group of PMAs between the PMD and the 800GAUI-8 closest to the PCS there shall be no more than 145 ns of Skew between PCSLs, and no more than 3.8 ns of Skew Variation between output lanes toward the 800GAUI-8 in the receive direction.

If there is a physically instantiated PMD service interface that allows the Skew to be measured, the Skew measured at SP5 is limited to no more than 127 ns of Skew and no more than 3.6 ns of Skew Variation. If there is no physically instantiated PMD service interface, the Skew measured at SP4 is limited to no more than 116 ns of Skew, and no more than 3.4 ns of Skew Variation.

#### 173.5.3.6 Skew tolerance at SP6

In an implementation with one or more physically instantiated 800GAUI-8 interfaces, the PMA between the 800GAUI-8 closest to the PCS and the PCS shall tolerate the maximum amount of Skew Variation allowed at SP6 (3.8 ns) between input lanes while maintaining the bit order and position of PCSLs on lanes sent in the receive direction towards the MAC.

### 173.5.4 Delay constraints

There may be up to four instances of the 800GBASE-R PMA within a Physical Layer, which is composed of an 800GBASE-R PHY and, optionally, an 800GMII Extender. The maximum delay (sum of transmit and receive delays) contributed by each instance of an 800GBASE-R PMA shall meet the values specified in Table 173–1. A description of overall system delay constraints and the definitions for bit times and pause\_quanta can be found in 169.4 and its references.

NOTE—Although the PMD service interface definition implies otherwise, the delay allocation for the majority of the clock and data recovery function is allocated to the PMA adjacent to the PMD, rather than to the PMD.

**Table 173–1—Delay constraints**

| Sublayer                        | Maximum<br>(bit time) | Maximum<br>(pause_quanta) | Maximum (ns) |
|---------------------------------|-----------------------|---------------------------|--------------|
| 800GBASE-R 8:32 PMA or 32:8 PMA | 36 864                | 72                        | 46.08        |
| 800GBASE-R 8:8 PMA              | 73 728                | 144                       | 92.16        |

### 173.5.5 Clocking architecture

A PMA with  $m$  input lanes and  $n$  output lanes shall clock the output lanes such that the bit rate of the output lanes is  $m/n$  times the bit rate of the input lanes. For input or output lanes encoded as PAM4 the symbol rate is half of the bit rate. This applies in both the transmit and receive directions of transmission.

NOTE—There is no requirement that all output lanes use a common clock. For example in an 8:8 PMA, each output lane may use a separate clock derived from its corresponding input.

Details of the clocking architecture are outside the scope of this standard.

### 173.5.6 Signal drivers

For a case where there is a physically instantiated interface, the PMA provides electrical signal drivers. The electrical and jitter/timing specifications for these interfaces appear in:

- Annex 120F, which specifies the 800GAUI-8 interface for chip-to-chip applications
- Annex 120G, which specifies the 800GAUI-8 interface for chip-to-module applications

### 173.5.7 PAM4 Encoding

#### 173.5.7.1 Gray mapping

The Gray mapping for PAM4 encoded lanes is identical to that specified in [120.5.7.1](#).

#### 173.5.7.2 Precoding

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of an 800GBASE-CR8 or 800GBASE-KR8 PMD, or are part of an 800GAUI-8 C2C link.

The PMA shall provide  $1/(1+D) \bmod 4$  precoding capability on each transmit lane and may optionally provide  $1/(1+D) \bmod 4$  decoding capability on each receive lane. Precoding is implemented as specified in [135.5.7.2](#).

The precoder is enabled independently on the Tx output, Rx input, Rx output, and Tx input on each lane. Precoding is enabled and disabled using variables `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei` (where  $i$  is in the range 0 to 7). If a Clause 45 MDIO is implemented, these variables are accessible through registers as shown in Table 173–2.

If the PMA is connected to the service interface of an 800GBASE-CR8 or 800GBASE-KR8 PMD and training is enabled by the management variable `mr_training_enable` (see [136.7](#)), then `precoder_tx_out_enablei` and `precoder_rx_in_enablei` shall be set as determined by the PMD control function in the LINK\_READY state on lane  $i$  (see [136.8.11.7.5](#) and [Figure 136–7](#)). The method by which the PMD control function affects these variables is implementation dependent.

If the PMA is connected to the service interface of an 800GBASE-CR8 or 800GBASE-KR8 PMD and training is disabled by the management variable `mr_training_enable`, or if the PMA is part of an 800GAUI-8 link, then `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei` are set as required by the implementation. The method described in [135F.3.2.1](#) may be used for 800GAUI-8 C2C.

## 173.5.8 Signal status

### 173.5.8.1 32:8 PMA signal status

In the receive direction, the 32:8 PMA provides signal status information to the PMA client (800GBASE-R PCS or DTE 800GXS) using the PMA:IS\_SIGNAL.indication(SIGNAL\_OK) service interface primitive (see 173.2 and Figure 173–3):

The SIGNAL\_OK parameter is set to OK when all of the following conditions are met:

- Data is being received on all 8 input lanes (*inst*:IS\_UNITDATA\_0:7.indication)
- The received data is being sent on all 32 output lanes (PMA:IS\_UNITDATA\_0:31.indication)
- The SIGNAL\_OK parameter of the PMD:IS\_SIGNAL.indication primitive is set to OK, if there is a PMD immediately below the PMA

Otherwise SIGNAL OK is set to FAIL.

### 173.5.8.2 8:32 PMA signal status

In the transmit direction, the 8:32 PMA provides signal status information to the PHY 800GXS using the PHY\_XS:IS\_SIGNAL.request(SIGNAL\_OK) service interface primitive (see 173.3 and Figure 173–4). The SIGNAL\_OK parameter is set to OK when data is being received on all 8 input lanes (PMA:IS\_UNITDATA\_0:7.request) and the received data is being sent on all 32 output lanes (PHY\_XS:IS\_UNITDATA\_0:31.request). Otherwise SIGNAL OK is set to FAIL.

In the receive direction, the 8:32 PMA optionally provides signal status information to the client PMA by disabling (squelching) one or more output lanes (PMA:IS\_UNITDATA\_0:7.indication) when the PHY\_XS:IS\_SIGNAL.indication SIGNAL\_OK parameter (see 173.3 and Figure 173–4) is FAIL.

### 173.5.8.3 8:8 PMA signal status

In the transmit direction, the 8:8 PMA optionally provides signal status to the sublayer below (see 173.3 and Figure 173–5) by disabling (squelching) one or more output lanes (*inst*:IS\_UNITDATA\_0:7.request) when data is not being received on all 8 input lanes (PMA:IS\_UNITDATA\_0:7.request).

In the receive direction, the 8:8 PMA optionally provides signal status to the client PMA by disabling (squelching) one or more output lanes (PMA:IS\_UNITDATA\_0:7.indication) when the PMD:IS\_SIGNAL.indication SIGNAL\_OK parameter (see 173.3 and Figure 173–5) is FAIL or when data is not being received on all 8 input lanes (*inst*:IS\_UNITDATA\_0:7.indication).

## 173.5.9 PMA local loopback mode (optional)

The PMA local loopback mode functions identically to that specified in 120.5.9.

## 173.5.10 PMA remote loopback mode (optional)

The PMA remote loopback mode functions identically to that specified in 120.5.10.

## 173.5.11 PMA test patterns (optional)

Where the lanes of the PMA are connected to a physically instantiated interface 800GAUI-8 or the PMD service interface, the PMA may optionally generate and detect test patterns. These test patterns are used to test adjacent layer interfaces for an individual PMA sublayer or to perform testing between a physically instantiated interface of a PMA and external testing equipment.

The test pattern functionality is identical to that defined for PAM4 encoded signals in 120.5.11.2.

## 173.6 PMA MDIO function mapping

The optional MDIO capability described in Clause 45 describes several variables that provide control and status information for and about the PMA. Since a given implementation may employ more than one PMA sublayer, the PMA control and status information is organized into multiple addressable instances, one for each possible PMA sublayer. See 45.2.1 and 173.1.4 for the allocation of MMD addresses to PMA sublayers. Control and status registers for MMD 8, 9, and 10 use the Extended PMA control and status registers at identical locations to those for MMD 1. Annex 173A provides additional examples of mapping MMD addresses to sublayers.

Mapping of MDIO control variables to PMA control variables is shown in Table 173–2. Mapping of MDIO status variables to PMA status variables is shown in Table 173–3. Mapping of MDIO counter to PMA counters is shown in Table 173–4. These tables provide the register and bit numbers for the PMA addressed as MMD 1. For implementations with multiple PMA sublayers, additional PMA sublayers use the corresponding register and bit numbers in MMDs 8, 9, 10, and 11 as necessary.

**Table 173–2—MDIO/PMA control variable mapping**

| MDIO variable                         | PMA/PMD register name          | Register/<br>bit number | PMA control variable         |
|---------------------------------------|--------------------------------|-------------------------|------------------------------|
| PMA remote loopback                   | PMA/PMD control 1              | 1.0.1                   | Remote_loopback_enable       |
| PMA local loopback                    | PMA/PMD control 1              | 1.0.0                   | Local_loopback_enable        |
| Lane 0 to 7 Tx output precoder enable | PMA precoder control Tx output | 1.600.0 to 1.600.7      | precoder_tx_out_enable_<0:7> |
| Lane 0 to 7 Rx input precoder enable  | PMA precoder control Rx input  | 1.601.0 to 1.601.7      | precoder_rx_in_enable_<0:7>  |
| Lane 0 to 7 Rx output precoder enable | PMA precoder control Rx output | 1.602.0 to 1.602.7      | precoder_rx_out_enable_<0:7> |
| Lane 0 to 7 Tx input precoder enable  | PMA precoder control Tx input  | 1.603.0 to 1.603.7      | precoder_tx_in_enable_<0:7>  |
| PRBS31Q pattern enable                | PRBS pattern testing control   | 1.1501.13               | PRBS31Q_pattern_enable       |
| SSPRQ pattern enable                  | PRBS pattern testing control   | 1.1501.14               | SSPRQ_pattern_enable         |
| Tx generator enable                   | PRBS pattern testing control   | 1.1501.3                | PRBS_Tx_gen_enable           |
| Tx checker enable                     | PRBS pattern testing control   | 1.1501.2                | PRBS_Tx_check_enable         |
| Rx generator enable                   | PRBS pattern testing control   | 1.1501.1                | PRBS_Rx_gen_enable           |
| Rx checker enable                     | PRBS pattern testing control   | 1.1501.0                | PRBS_Rx_check_enable         |
| Lane 0 to 7 SW enable                 | Square wave testing control    | 1.1510.0 to 1.1510.7    | Square_wave_enable_<0:7>     |
| Lane 0 to 7 PRBS9Q enable             | PRBS9Q testing control         | 1.1511.0 to 1.1511.7    | PRBS9Q_enable_<0:7>          |
| Lane 0 to 7 PRBS13Q enable            | PRBS13Q testing control        | 1.1512.0 to 1.1512.7    | PRBS13Q_enable_<0:7>         |

**Table 173–3—MDIO/PMA status variable mapping**

| MDIO status variable                         | PMA/PMD register name                  | Register/bit number | PMA status variable          |
|----------------------------------------------|----------------------------------------|---------------------|------------------------------|
| PMA remote loopback ability                  | 800G PMA/PMD extended ability register | 1.73.15             | 800G_Remote_loopback_ability |
| PMA local loopback ability                   | PMA/PMD status 2 register              | 1.8.0               | Local_loopback_ability       |
| Tx input precoder request flag               | PMA precoder request flag              | 1.604.1             | precoder_rx_in_request_flag  |
| Rx input precoder request flag               | PMA precoder request flag              | 1.604.0             | precoder_rx_in_request_flag  |
| Lane 0 to 7 Rx input precoder request status | PMA precoder request Rx input status   | 1.605.0 to 1.605.7  | precoder_rx_in_request_<0:7> |
| Lane 0 to 7 Tx input precoder request status | PMA precoder request Tx input status   | 1.606.0 to 1.606.7  | precoder_tx_in_request_<0:7> |
| Square wave test ability                     | Test-pattern ability register          | 1.1500.12           | Square_wave_ability          |
| PRBS13Q Tx generator ability                 | Test-pattern ability register          | 1.1500.11           | PRBS13Q_gen_Tx_ability       |
| PRBS13Q Rx generator ability                 | Test-pattern ability register          | 1.1500.10           | PRBS13Q_gen_Rx_ability       |
| PRBS31Q Tx generator ability                 | Test-pattern ability register          | 1.1500.9            | PRBS31Q_gen_Tx_ability       |
| PRBS31Q Rx generator ability                 | Test-pattern ability register          | 1.1500.7            | PRBS31Q_gen_rx_ability       |
| PRBS31Q Tx checker ability                   | Test-pattern ability register          | 1.1500.8            | PRBS31Q_Tx_checker_ability   |
| PRBS31Q Rx checker ability                   | Test-pattern ability register          | 1.1500.6            | PRBS31Q_Rx_checker_ability   |
| SSPRQ Tx generator ability                   | Test-pattern ability register          | 1.1500.13           | SSPRQ_gen_Tx_ability         |
| PRBS9Q Rx generator ability                  | Test-pattern ability register          | 1.1500.14           | PRBS9Q_gen_Rx_ability        |
| PRBS9Q Tx generator ability                  | Test-pattern ability register          | 1.1500.15           | PRBS9Q_gen_Tx_ability        |

**Table 173–4—MDIO/PMA counters mapping**

| MDIO variable                 | PMA/PMD register name                                   | Register/bit number | PMA status variable              |
|-------------------------------|---------------------------------------------------------|---------------------|----------------------------------|
| Error counter Tx, lane 0 to 7 | PRBS Tx pattern testing error counter, lane 0 to lane 7 | 1.1600 to 1.1607    | Ln<0:7>_PRBS_Tx_test_err_counter |
| Error counter Rx, lane 0 to 7 | PRBS Rx pattern testing error counter, lane 0 to lane 7 | 1.1700 to 1.1707    | Ln<0:7>_PRBS_Rx_test_err_counter |

## 173.7 Protocol implementation conformance statement (PICS) proforma for Clause 173, Physical Medium Attachment (PMA) sublayer, type 800GBASE-R<sup>14</sup>

### 173.7.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 173, Physical Medium Attachment (PMA) sublayer, type 800GBASE-R shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 173.7.2 Identification

#### 173.7.2.1 Implementation identification

|                                                                                                                                                                                                                                                                                          |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                                                                                                                                                                    |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                                                                                                                                                                  |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                                                                                                                                                     |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup>                                                                                                                                      |  |
| NOTE 1—Required for all implementations.<br>NOTE 2—May be completed as appropriate in meeting the requirements for the identification.<br>NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model). |  |

#### 173.7.2.2 Protocol summary

|                                                                                                                                                                    |                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                | IEEE Std 802.3df-2024, Clause 173, Physical Medium Attachment (PMA) sublayer, type 800GBASE-R |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                    |                                                                                               |
| Have any Exception items been required? No [ ] Yes [ ]<br>(See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.) |                                                                                               |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>14</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

### 173.7.3 Major capabilities/options

| Item  | Feature                                                                               | Subclause | Value/Comment                           | Status       | Support                      |
|-------|---------------------------------------------------------------------------------------|-----------|-----------------------------------------|--------------|------------------------------|
| *P328 | 32:8 PMA                                                                              | 173.4     |                                         | O/1          | Yes [ ]<br>No [ ]            |
| *P832 | 8:32 PMA                                                                              | 173.4     |                                         | O/1          | Yes [ ]<br>No [ ]            |
| *P88  | 8:8 PMA                                                                               | 173.4     |                                         | O/1          | Yes [ ]<br>No [ ]            |
| *C2CA | AUI C2C above PMA<br>(toward MAC)                                                     | 173.2     |                                         | P832+P88:O/2 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *C2MA | AUI C2M above PMA<br>(toward MAC)                                                     | 173.2     |                                         | P832+P88:O/2 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *C2CB | AUI C2C below PMA<br>(toward PMD)                                                     | 173.3     |                                         | P328+P88:O/3 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *C2MB | AUI C2M below PMA<br>(toward PMD)                                                     | 173.3     |                                         | P328+P88:O/3 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PMDE | 800GBASE-CR8 or<br>800GBASE-KR8<br>PMD below PMA                                      | 173.3     |                                         | P328+P88:O/3 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PMDO | 800GBASE-VR8,<br>800GBASE-SR8,<br>800GBASE-DR8, or<br>800GBASE-DR8-2<br>PMD below PMA | 173.3     |                                         | P328+P88:O/3 | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PSID | PMA connected to a<br>physically instantiated<br>PMD service interface                | 173.3     |                                         | O            | Yes [ ]<br>No [ ]            |
| *MD   | MDIO                                                                                  | 173.6     | Registers and<br>interface<br>supported | O            | Yes [ ]<br>No [ ]            |

### 173.7.4 Skew generation and tolerance

| Item | Feature                                                       | Subclause | Value/Comment  | Status      | Support            |
|------|---------------------------------------------------------------|-----------|----------------|-------------|--------------------|
| S1   | Skew generation toward SP1 in<br>transmit direction           | 173.5.3.1 | ≤ 16 ns        | C2CB+C2MB:M | Yes [ ]<br>N/A [ ] |
| S2   | Skew Variation generation<br>toward SP1 in transmit direction | 173.5.3.1 | ≤ 200 ps       | C2CB+C2MB:M | Yes [ ]<br>N/A [ ] |
| S3   | Skew Variation tolerance at SP1                               | 173.5.3.2 | Minimum 200 ps | C2CA+C2MA:M | Yes [ ]<br>N/A [ ] |
| S4   | Skew generation toward SP2 in<br>transmit direction           | 173.5.3.3 | ≤ 25 ns        | PSID:M      | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                    | Subclause | Value/Comment  | Status      | Support            |
|------|------------------------------------------------------------|-----------|----------------|-------------|--------------------|
| S5   | Skew Variation generation toward SP2 in transmit direction | 173.5.3.3 | $\leq 400$ ps  | PSID:M      | Yes [ ]<br>N/A [ ] |
| S6   | Skew Variation tolerance at SP5                            | 173.5.3.4 | Minimum 3.6 ns | PSID:M      | Yes [ ]<br>N/A [ ] |
| S7   | Skew generation toward SP6 in receive direction            | 173.5.3.5 | $\leq 145$ ns  | C2CA+C2MA:M | Yes [ ]<br>N/A [ ] |
| S8   | Skew Variation generation toward SP6 in receive direction  | 173.5.3.5 | $\leq 3.8$ ns  | C2CA+C2MA:M | Yes [ ]<br>N/A [ ] |
| S9   | Skew Variation tolerance at SP6                            | 173.5.3.6 | Minimum 3.8 ns | C2CB+C2MB:M | Yes [ ]<br>N/A [ ] |

### 173.7.5 Delay constraints

| Item | Feature              | Subclause | Value/Comment | Status | Support |
|------|----------------------|-----------|---------------|--------|---------|
| D1   | PMA delay constraint | 173.5.4   |               | M      | Yes [ ] |

### 173.7.6 Test patterns

| Item | Feature                          | Subclause    | Value/Comment | Status                    | Support                      |
|------|----------------------------------|--------------|---------------|---------------------------|------------------------------|
| T1   | Send PRBS13Q Tx                  | 120.5.11.2.1 |               | C2CB+C2MB+<br>PMDO+PMDE:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T2   | Send PRBS13Q Rx                  | 120.5.11.2.1 |               | C2CA+C2MA:O               | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T3   | Send PRBS31Q Tx                  | 120.5.11.2.2 |               | C2CB+C2MB+<br>PMDO+PMDE:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T4   | Send PRBS31Q Rx                  | 120.5.11.2.2 |               | C2CA+C2MA:O               | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T5   | Check PRBS31Q Tx                 | 120.5.11.2.2 |               | C2CA+C2MA:O               | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T6   | Check PRBS31Q Rx                 | 120.5.11.2.2 |               | C2CB+C2MB+<br>PMDO+PMDE:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T7   | Send SSPRQ Tx                    | 120.5.11.2.3 |               | PMDO:O                    | Yes [ ]<br>No [ ]<br>N/A [ ] |
| T8   | Send square wave (quaternary) Tx | 120.5.11.2.4 |               | PMDO:O                    | Yes [ ]<br>No [ ]<br>N/A [ ] |

### 173.7.7 Loopback

| Item | Feature             | Subclause | Value/Comment                      | Status | Support           |
|------|---------------------|-----------|------------------------------------|--------|-------------------|
| LB1  | PMA local loopback  | 173.5.9   | Meets the requirements of 173.5.9  | O      | Yes [ ]<br>No [ ] |
| LB2  | PMA remote loopback | 173.5.10  | Meets the requirements of 173.5.10 | O      | Yes [ ]<br>No [ ] |

### 173.7.8 Precoding

| Item | Feature                                | Subclause | Value/Comment | Status          | Support                      |
|------|----------------------------------------|-----------|---------------|-----------------|------------------------------|
| P1   | PMA transmit output supports precoding | 173.5.7.2 |               | PMDE+<br>C2CB:M | Yes [ ]<br>N/A [ ]           |
| P2   | PMA receive output supports precoding  | 173.5.7.2 |               | C2CA:M          | Yes [ ]<br>N/A [ ]           |
| P3   | PMA transmit input supports precoding  | 173.5.7.2 |               | C2CA:O          | Yes [ ]<br>No [ ]<br>N/A [ ] |
| P4   | PMA receive input supports precoding   | 173.5.7.2 |               | PMDE+<br>C2CB:O | Yes [ ]<br>No [ ]<br>N/A [ ] |

### 173.7.9 Signal status

| Item | Feature                                                           | Subclause | Value/Comment | Status | Support                      |
|------|-------------------------------------------------------------------|-----------|---------------|--------|------------------------------|
| SS1  | Signal status for 32:8 PMA                                        | 173.5.8.1 |               | P328:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| SS2  | Signal status for 8:32 PMA                                        | 173.5.8.2 |               | P832:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| SS3  | Signal status for 8:8 PMA                                         | 173.5.8.3 |               | P88:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| SS4  | Signal status by disabling (squenching) output lanes for 8:32 PMA | 173.5.8.2 |               | P832:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| SS5  | Signal status by disabling (squenching) output lanes for 8:8 PMA  | 173.5.8.3 |               | P88:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |

### 173.7.10 Electrical

| Item | Feature                                                        | Subclause           | Value/Comment | Status | Support            |
|------|----------------------------------------------------------------|---------------------|---------------|--------|--------------------|
| E1   | Interface above meets 800GAUI-8 C2C requirements in Annex 120F | 173.5.1,<br>173.5.6 |               | C2CA:M | Yes [ ]<br>N/A [ ] |
| E2   | Interface above meets 800GAUI-8 C2M requirements in Annex 120G | 173.5.1,<br>173.5.6 |               | C2MA:M | Yes [ ]<br>N/A [ ] |
| E3   | Interface below meets 800GAUI-8 C2C requirements in Annex 120F | 173.5.1,<br>173.5.6 |               | C2CB:M | Yes [ ]<br>N/A [ ] |
| E4   | Interface below meets 800GAUI-8 C2M requirements in Annex 120G | 173.5.1,<br>173.5.6 |               | C2MB:M | Yes [ ]<br>N/A [ ] |